

SEBASTIAN KOCH

Doctoral Researcher (final year)

✉ koch.sebastian@mail.com

🌐 <https://kochsebastian.com>

🇨🇭 Zurich, Switzerland

Education

Apr 2022 –
Dec 2026*

Ulm University, Germany

PhD in Computer Science

- *Advisor:* Prof. Timo Ropinski
- *Thesis:* Compositional and Language-guided 3D Scene Understanding
- *Apr 2022 – Apr 2025:* Bosch Research Doctoral Program
- *May 2025 – Nov 2025:* Google Student Researcher
- *Mar 2026 – Jul 2026:* Google Research Intern

2020 – 2022

Tübingen University, Germany

Master of Science in Computer Science

- *Advisors:* Prof. Gerhard Neumann & Prof. Andreas Geiger
- *Thesis project:* Multi-View RGB-D Fusion for 6D Pose Estimation
- *GPA:* 1.4 (Scale: 1.0 [best] to 5.0 [fail])
- *Selected courses:* Computer Vision, Deep Learning, 3D Computer Vision, Mobile Robotics, Reinforcement Learning, Mathematics for Machine Learning

2016 – 2020

Baden-Württemberg Cooperative State University (DHBW), Germany

Bachelor of Engineering in Computer Science

- *Thesis project:* Improvement of the robustness of a SLAM system using Object Semantics
- *GPA:* 1.8 (Scale: 1.0 [best] to 5.0 [fail])

Experience

Google Research, Zurich Switzerland

Mar 2026 –
Jul 2026

Research Intern – 3D Scene Reconstruction and Generation for Android XR

- *Advisors:* Marie-Julie Rakotosaona & Federico Tombari
- *Role:* Research on the joint benefits of 3D reconstruction and world models with 3D Gaussian Splatting and video generation models.

Google Research, Munich Germany

May 2025 –
Nov 2025

Student Researcher – 3D Scene Understanding for AR/VR Applications

- *Advisors:* Johanna Wald & Federico Tombari
- *Role:* Research on open-vocabulary 3D scene understanding using forward models, investigating object semantics, affordances & articulations [1]. Built up a forward model implementation that is used internally across multiple research projects.

Bosch Center for Artificial Intelligence Stuttgart Germany

Apr 2022 –
Apr 2025

Bosch Research Doctoral Program – 3D Scene Understanding for indoor Robotic Applications

- *Advisors:* Mirco Colosi & Narunas Vaskevicius
- *Role:* Research and development on 3D Scene Graphs for indoor 3D scene understanding, achieving over 50% reduction in labeled sample requirements through self-supervised learning, and introducing open-vocabulary relationship predictions for fine-grained inter-object reasoning. This work resulted in four first-author publications [3], [5], [6], [7].

Oct 2021 –
Mar 2022

Industrial Master Thesis – 6D Pose Estimation

- *Advisor:* Fabian Duffhauss

- **Role:** Developed a multi-view RGB-D fusion method with a symmetry-aware keypoint voting approach, improving 6D Pose Estimation accuracy by 7.6% and achieving state-of-the-art results, resulting in a publication [8].

University of Tübingen, Tübingen Germany

Sep 2020 –
Oct 2021

Research Assistant – Real-time high-resolution remote sensing

- **Advisors:** Leon-Amadeus Varga & Prof. Andreas Zell
- **Role:** Research on encoder-only region proposal techniques combined with hardware-optimized TensorRT and CUDA implementations on embedded GPUs for real-time object detection in high-resolution images. Studied the impact of on-device image pre-processing for enhanced remote sensing object detection accuracy, resulting in a publication [9].

Bosch Research, Stuttgart Germany

Apr 2020 –
Oct 2020

Research Intern – Object-level Semantics for SLAM

- **Advisor:** Stefan Benz
- **Role:** Integrated an object detection pipeline into a ROS system for enhanced localization and mapping. Led synthetic data generation using Unreal Engine to enable reproducible mapping runs used in the evaluation section of a published paper.

Bosch Group, Stuttgart Germany

Oct 2016 –
Mar 2020

Industrial Bachelor Student

- **Role:** Alongside my bachelor's in computer science at the university, I worked on scientific projects in different departments at Bosch, applying learned knowledge and collaborating with experienced developers and researchers.
- **Mentors:** Stefan Benz, Hanna Ziesche, Christopher Baker, Stephan Stühmer
- **Nov 2017 – Feb 2018:** Analyzed false-positive ultrasonic reflections in automotive sensors.
- **Jun 2018 – Sep 2018:** Implemented a play-back recording feature for ECU failure analysis.
- **Jun 2019 – Sep 2019:** Implemented a pedestrian simulator for ML-based human behavior prediction.
- **Sep 2019 – Mar 2020:** Researched the impact of semantic features for improved SLAM accuracy.

Honors & Awards

- 2024 | Accepted into International Computer Vision Summer School (ICVSS) 2024 for excellent PhDs in CV.
- 2021 | 1st place in the AI Chess Competition conducted by the Cognitive Systems Lab of Prof. Andreas Zell.
3rd place in the RL Hockey Competition of the MPI Autonomous Learning Group of Prof. Georg Martius.
- 2020 | Accepted into the Students@Bosch program for students who excelled at Bosch internships.
- 2014 | 4th place at the RoboCup World Cup 2014 in Joao Pessoa Brazil in the Rescue Junior competition
- 2013 | German Champion at the RoboCup German Open 2013 in the Rescue Junior competition.
World Champion at the RoboCup World Cup 2013 in Eindhoven in the Rescue Junior competition.

Publications

For a complete list of all publications, including recent pre-preprints see: kochsebastian.com/publications

- 2025 [1] **S Koch**, J Wald, H Matsuki, P Hermosilla, T Ropinski, F Tombari, “*Unified Semantic Transformer for 3D Scene Understanding*”, arXiv:2512.14364, 2025.
- [2] L Weijler, **S Koch**, F Poiesi, T Ropinski, P Hermosilla, “*OpenHype: Hyperbolic Embeddings for Hierarchical Open-Vocabulary Radiance Fields*”, in Advances in Neural Information Processing Systems (**NeurIPS**), 2025.
- [3] **S Koch**, J Wald, M Colosi, N Vaskevicius, P Hermosilla, F Tombari, T Ropinski, “*RelationField: Relate Anything in Radiance Fields*”, in Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025.

- [4] Y Liu, L Palmieri, **S Koch**, I Georgievski, M Aiello, “*DELTA: Decomposed Efficient Long-Term Robot Task Planning using Large Language Models*”, in IEEE International Conference on Robotics and Automation (**ICRA**), 2025.
- 2024 [5] **S Koch**, N Vaskevicius, M Colosi, P Hermosilla, T Ropinski, “*Open3DSG: Open-vocabulary 3D Scene Graphs from Point Clouds with Queryable Objects and Open-Set Relationships*”, in Conference on Computer Vision and Pattern Recognition (**CVPR**), 2024.
- [6] **S Koch**, P Hermosilla, N Vaskevicius, M Colosi, T Ropinski, “*Lang3DSG: Language-based contrastive pre-training for 3D Scene Graph prediction*”, in International Conference on 3D Vision (**3DV**), 2024.
- [7] **S Koch**, P Hermosilla, N Vaskevicius, M Colosi, T Ropinski, “*SGRec3D: Self-Supervised 3D Scene Graph Learning via Object-Level Scene Reconstruction*”, in Winter Conference on Applications of Computer Vision (**WACV**), 2024.
- 2023 [8] F Duffhauss, **S Koch**, H Ziesche, NA Vien, G Neumann, “*SyMFM6D: Symmetry-aware Multi-directional Fusion for Multi-View 6D Object Pose Estimation*”, in Robotics and Automation Letters (**RA-L**), 2023.
- 2022 [9] LA Varga, **S Koch**, A Zell, “*Comprehensive Analysis of the Object Detection Pipeline on UAVs*”, in **Remote Sensing**, 2022.

Invited Talks

- 2025 **Google Research: Open-Vocabulary Relationship Reasoning in 3D**, Munich 05.05.2025.
Huawei Munich Research Center: Language-driven Scene Understanding with 3D Scene Graphs, Munich 06.02.2025.
- 2024 **RWTH Aachen. Computer Vision Group, Prof. Bastian Leibe: 3D Scene Understanding with label-efficient and open-vocabulary 3D Scene Graphs**, Remote 27.02.2024.
Bosch Research US, Sunnyvale. 3D Scene Understanding with label-efficient and open-vocabulary 3D Scene Graphs, Remote 07.02.2024.

Service

Reviewing: Conferences: CVPR 2024–2026, ICCV 2023–2025, NeurIPS 2024–2025, ICML 2026, ICRA 2025, ROS 2024–2025 — Journals: RA-L 2024–2026 — Recognition: Outstanding Reviewer: CVPR 2025
Volunteering: I work as a volunteer and referee at RoboCup Junior events on a national and international level.

Qualifications

Programming: Python, C/C++, MATLAB, Bash
Libraries: PyTorch, JAX, NumPy, Lightning, FLAX, Matplotlib, Open3D, OpenCV, TensorRT
Software: Git, $\text{\LaTeX}$, ROS, GNU/Linux, Blender, Office
Languages: German (native), English (fluent)

References

Prof. Timo Ropinski. Full Professor, Ulm University.	timo.ropinski@uni-ulm.de
Dr. Federico Tombari. Director of Research, Google Zurich.	tombari@google.com
Prof. Pedro Hermosilla. Assistant Professor, TU Wien.	phermosilla@cvl.tuwien.ac.at
Dr. Narunas Vaskevicius. Lead Research Scientist, Bosch Research.	narunas.vaskevicius@bosch.com
Prof. Gerard Pons-Moll. Full Professor, University of Tuebingen.	gerard.pons-moll@uni-tuebingen.de